DOCUMENT CONTROL

Document ID	EAAP-ARCH-003A
Title	Orchestration Decision Model:Production Design
Version	v1.0
Status	Production Design
Classification	Internal / Controlled
Parent	EAAP-ARCH-003
Owner	John Goulden
Date	02.04.2026
Contact	research@johngoulden.de

Version History

Version / Date	Author	Description
v1.0 | 02.04.2026	John Goulden	Initial release. ID corrected from EAAP-ARCH-003. All placeholder sections populated.

Table of Contents

1. ORCHESTRATION DECISION MODEL: PRODUCTION DESIGN

1.1 Purpose and Objectives

This document defines the production design of the EAAP Orchestration Decision Model as implemented in the EAAP platform. It translates the abstract model in EAAP-ARCH-003 into specific operational design decisions, component mappings, and production constraints.

- **Define the production orchestrator component architecture**
- **Specify the evidence framework for the current implementation**
- **Document failure handling behaviour and ITSM integration**
- **Record security controls and audit requirements for the production deployment**

1.2 Scope

This document covers the production design of orchestration decision logic within the EAAP platform: orchestrator components, decision logic, evidence classification, failure handling, ITSM mapping, security controls, and audit requirements.

Excluded: abstract architectural principles (EAAP-ARCH-003), domain agent internal design (EAAP-ARCH-001), deployment infrastructure (EAAP-ARCH-006).

1.3 Orchestration Principles

- **Deterministic execution only** One recommendation per invocation based on evidence and rules.
- **Structured input/output enforcement** All inputs and outputs conform to defined schemas.
- **Evidence-based reasoning mandatory** Every output linked to an evidence class.
- **Human approval for write actions** No write action without explicit human approval token.

1.4 Orchestrator Architecture

Component	Function	Implementation
Request Classifier	Determines intent type.	Prompt-based classification against six intent types.
Domain Router	Maps request to domain agents.	Rule-based routing against domain authority matrix.
Execution Planner	Defines agent invocation sequence.	Sequential or parallel based on dependency type.
Agent Executor	Invokes domain agents and collects outputs.	System prompt injection. Outputs as structured JSON.
Evidence Aggregator	Normalises and classifies evidence.	Evidence hierarchy applied. Lowest class governs.
Conflict Resolver	Detects and resolves conflicts.	Four-level resolution hierarchy from EAAP-ARCH-003.
Output Composer	Builds the final structured output.	Mandatory 15-field structure from EAAP-ARCH-003.

1.5 Decision Model

The production decision model implements the core flow from EAAP-ARCH-003 with the following enforcement gates:

- **Data Sufficiency Gate** Before domain invocation. Insufficient data aborts the flow.
- **Confidence Gate** Before output composition. Low/Unknown confidence triggers Conditional Override.
- **Escalation Gate** Before output delivery. Escalation conditions halt delivery.
- **Governance Validation** After output composition. Output validated before delivery.

1.6 Evidence Framework

Level	Description	Example
Vendor-Supported	Official vendor documentation or support articles.	Nutanix AOS 6.5 upgrade guide.
Framework-Supported	EAAP principles or referenced enterprise standards.	ISO 27001 access control requirements.

Best Practice	Industry-accepted practice with published source.	ITIL change management best practice.
Heuristic	Model inference. No direct documentary support.	General infrastructure sizing heuristic.
Assumption	Explicit assumption in absence of evidence.	Assumed network bandwidth adequate.

1.7 Failure Handling

Failure Type	Detection	Action
Data Insufficient	Data Sufficiency Gate fails.	Abort. Return missing data list. No output.
Agent Failure	Domain agent returns no structured output.	Retry once. If retry fails, mark domain unavailable.
Conflict Unresolved	Conflict Resolver cannot apply hierarchy.	Escalate. Return conflict detail. No recommendation.
Execution Failure	Gateway returns execution error.	Block. Log error. Raise ITSM incident.
Confidence Too Low	Confidence Gate returns Low/Unknown.	Conditional Override. Return data requirements.

1.8 ITSM Integration

EAAP Step	ITSM Object	Required Fields
Plan	Change Request	Change ID, scope, blast radius, rollback plan, maintenance window.
Approval	CAB Decision	Approval token, CAB reference, approver identity, expiry timestamp.
Execution	Change Task	Idempotency key, plan ID, approval token, execution log reference.
Failure	Incident	Incident ID, failure type, affected systems, remediation steps.

1.9 Security Controls

- **RBAC per agent** Keycloak OIDC tokens scoped per domain agent capability.
- **Read/write scope separation** Read tokens cannot invoke write actions. Separate scopes enforced at Gateway.
- **Token-based approval** Tier 2 and Tier 3 actions require a time-limited approval token.
- **Data isolation** Each domain agent receives only the data required for its invocation.
- **Audit logging** Every orchestration invocation logged with user identity, agents invoked, confidence, conflicts, and escalation status.
- **Prompt data protection** Prompt content never logged in cleartext. Hashed reference only.

1.10 Audit and Traceability

Every orchestration decision produces an immutable audit record containing:

- **User identity** Authenticated principal from Keycloak token.
- **Agents invoked** List with invocation timestamps.
- **Intent classification** Classified intent type.
- **Conflict log** Conflicts detected and resolution applied.
- **Confidence assignment** Overall and per-domain confidence levels.
- **Escalation status** Whether escalation was triggered and why.
- **Output reference** Reference ID linking audit record to delivered output.

1.11 Implementation Model

Layer	Technology
Inference	Ollama serving LLaMA 3 (pinned version) on Nutanix AHV
Orchestrator	EAAP system prompt layer (EAAP proprietary IP)

UI	Open WebUI (containerised, RBAC-enabled)
Container Runtime	Docker (rootless, pinned digests) on RHEL 8/9
IAM	Keycloak (OIDC, RBAC)
Knowledge / RAG	Obsidian corpus with FAISS or Qdrant vector index
Automation Gateway	FastAPI with Celery/Redis task queue
Audit Logging	OpenSearch (immutable, write-once index)
Version Control	Gitea (local, air-gapped)

1.12 Risks and Assumptions

Risks

- **Orchestrator complexity** Increasing domain count increases routing complexity. Mitigate: maintain domain authority matrix.
- **Data quality dependency** Output quality bounded by input data quality. Mitigate: enforce Data Sufficiency Gate.
- **Agent statefulness** Agents must remain stateless. Mitigate: architectural enforcement.

Assumptions

- **Domain agents conform to EAAP-ARCH-001 structural requirements.**
- **Knowledge base is current and covers the domains being orchestrated.**
- **ITSM integration available for Tier 2 and Tier 3 action approval.**

1.13 Confidence Level

Document Confidence	High
Basis	Production deployment validated. All components operational.